Spatial SNR Calculation for Single Molecular Imaging
单分子成像的信噪比 (?) 计算

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1 General Theory

Random noise with definite signal

To calculate the SNR of a given signal, we need to identify first what signal and noise are:

Definition 1 (SNR)**.**

$$SNR = \frac{I_{signal}}{I_{noise}} \tag{1}$$

Generally, our object may be represented as a function $f(\vec{x})$, from $\mathbb{E} \to \mathbb{R}$. Often, the noise is random so we must recognize it as a random variable I_{noise}, who gives different value each experiment. Now, if we define **what will always exists without the signal** as noise, and suppose a **linearity** of system(eg. Imaging), then value observed in experiments with a signal existing should be:

$$I_{exp} = I_{noise} + I_{signal} \tag{2}$$

Thus the expected ones will be:

$$\mathbb{E}I_{exp} = \mathbb{E}I_{noise} + I_{signal} \tag{3}$$

So when signal is Constant, we may subtract averaged noise(measured with out signal) from I_{exp} to get a relative value whose expection is true signal I_{real}:

$$I_{rel} \triangleq I_{exp} - \overline{I_{noise}} \tag{4}$$

$$\mathbb{E}I_{rel} = I_{signal} \tag{5}$$

Now that we get the relative value, we further recognize that random noise will fluctuate around its expectation. If the extent of fluctuation is comparable to the relative value, it is expected that the signal may be distorted. So we use the variance to serve as noise in the formula:

$$SNR = \frac{I_{rel}}{\sigma_{noise}} \tag{6}$$

Random noise with random signal

Now that the signal itself is random, we must think of the true one as the expectation of signal.

$$\mathbb{E}I_{exp} = \mathbb{E}I_{noise} + \mathbb{E}I_{signal} \tag{7}$$

We still need to subtract averaged noise from observed value, but now I_{rel} will certainly have **more fluctuation** then before. The problem is how can we estimate the variance of the signal itself.

If we can get rid of the background noise, we may calculate the variance over experiments. But that's not a problem, since the variance is linear over 2 random variables. The key is to estimate the expectation of I_{rel}:

$$\sigma_{total}^2 = \sigma_{noise}^2 + \sigma_{signal}^2 \tag{8}$$

$$\sigma_{total}^2 = \mathbb{E}[(I_{rel} - \mathbb{E}I_{rel})^2] \tag{9}$$

We write this conclusion into our formula:

$$SNR = \frac{\mathbb{E}I_{rel}}{\sigma_{total}} \tag{10}$$

2 Analysis in smImaging

Theoretical variance of sm